

Database Schema Specification

Complete Relational Data Model for the Techvunex AI Customer Assistant, Lead Capture Pipeline, Real-Time Conversations, and Hybrid RAG Knowledge Engine.

8

CORE TABLES

PostgreSQL /
SQLite

SUPPORTED DRIVERS

AsyncIO

SQLALCHEMY 2.0

Hybrid RAG

VECTOR & BM25 READY

Entity Relationship Architecture



1 Lead Generation & Main Website Sync Tables

Leads Captures prospective customer inquiries, project details, budgets, and qualification status.

PRIMARY CRM TABLE

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Globally unique lead identifier
name	VARCHAR(100)	NULL	NULL	Prospect's full name
email	VARCHAR(150)	INDEX	NULL	Email address for follow-ups
phone	VARCHAR(30)	NULL	NULL	10-digit phone / WhatsApp number
company	VARCHAR(150)	NULL	NULL	Business or enterprise name
service	VARCHAR(100)	NULL	NULL	Requested service (e.g. Website, CRM, AI)
requirement	TEXT	NULL	NULL	Project scope discussed in chat
budget	VARCHAR(100)	NULL	NULL	Customer's stated budget bracket
timeline	VARCHAR(100)	NULL	NULL	Target launch date / urgency
status	VARCHAR(30)	INDEX	'new'	'new', 'contacted', 'won', 'human_required'
source	VARCHAR(50)	NULL	'website_chat'	Lead source channel
human_required	BOOLEAN	NULL	FALSE	Flag if human callback requested
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Lead creation timestamp (UTC)
updated_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Last modification timestamp (UTC)

contacts Main web portal synchronized table for direct contact submissions.

PORTAL SYNC

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	INTEGER	PK AUTO	AUTO_INCREMENT	Sequential integer identifier
name	VARCHAR(255)	NULL	NULL	Full contact name
email	VARCHAR(255)	NULL	NULL	Contact email
phone	VARCHAR(50)	NULL	NULL	Contact phone number
company	VARCHAR(255)	NULL	NULL	Company / Organization
subject	VARCHAR(255)	NULL	NULL	Message subject / Inquired service
message	TEXT	NULL	NULL	Enquiry message / requirements
createdAt	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Submission timestamp
updatedAt	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Update timestamp

2 Real-Time Conversation & Message Audit Tables

conversations Stores user sessions, dialog intents, and conversation lifecycles.

SESSION ROOT

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Internal conversation UUID
session_id	VARCHAR(100)	UNIQUE INDEX	NOT NULL	Client-side widget session key
user_id	VARCHAR(36)	FK	NULL	References users.id (ON DELETE SET NULL)
title	VARCHAR(255)	NULL	'New Conversation'	Descriptive title of session
intent	VARCHAR(50)	NULL	'general_question'	Classified conversational intent
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Session creation timestamp
updated_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Last interaction timestamp

messages Maintains complete transcript of all turns between user, assistant, and system.

CHAT HISTORY

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Message UUID
conversation_id	VARCHAR(36)	FK NOT NULL	CASCADE DELETE	References conversations.id
role	VARCHAR(20)	NOT NULL	—	'user', 'assistant', or 'system'
content	TEXT	NOT NULL	—	Message text (supports Markdown)
sources	JSON	NULL	[]	RAG citation references & URLs
token_usage	JSON	NULL	{}	Prompt, completion & total tokens
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Message creation timestamp

3 Quality Feedback & User Authentication

feedbacks User quality ratings (thumbs up/down) for continuous model alignment.

QUALITY RATING

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Feedback UUID
conversation_id	VARCHAR(36)	FK NOT NULL	CASCADE DELETE	References conversations.id
message_id	VARCHAR(36)	NULL	NULL	Optional targeted message ID
rating	INTEGER	NOT NULL	—	+1 (Helpful 👍) or -1 (Not helpful 👎)
feedback	TEXT	NULL	NULL	User review comment
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Rating submission timestamp

users Authentication and profiles for system administrators and registered users.

AUTH ENTITY

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	User UUID
name	VARCHAR(100)	NULL	NULL	User full name
email	VARCHAR(150)	UNIQUE INDEX	NULL	Account login email
phone	VARCHAR(30)	NULL	NULL	Contact phone number
company	VARCHAR(150)	NULL	NULL	Company affiliation
role	VARCHAR(20)	NULL	'visitor'	'admin' or 'visitor'
password_hash	VARCHAR(255)	NULL	NULL	Argon2 / bcrypt password hash
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Account registration timestamp

4 RAG Knowledge Base & Neural Vector Index Tables

documents Master store for crawled Techvunex web pages and company knowledge docs.

KNOWLEDGE ROOT

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Document UUID
url	VARCHAR(500)	UNIQUE INDEX	NOT NULL	Source URL on techvunex.in
title	VARCHAR(255)	NOT NULL	—	Page or section heading
content	TEXT	NOT NULL	—	Full raw markdown / text content
content_hash	VARCHAR(64)	NOT NULL	—	SHA-256 hash for change detection
metadata_info	JSON	NULL	{}	SEO tags, crawl timestamps, category
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Ingestion timestamp
updated_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Last refresh timestamp

document_chunks Semantic text chunks and dense vector embeddings used by hybrid search.

VECTOR STORE

COLUMN NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
id	VARCHAR(36)	PK	UUID v4	Chunk UUID
document_id	VARCHAR(36)	FK NOT NULL	CASCADE DELETE	References documents.id
chunk_index	INTEGER	NOT NULL	—	Sequential chunk order in document
content	TEXT	NOT NULL	—	Chunk text payload (~700 chars)
embedding	JSON	NULL	NULL	384-dimensional dense vector array
metadata_info	JSON	NULL	{}	Section name, title, keywords
created_at	TIMESTAMP	NULL	CURRENT_TIMESTAMP	Chunk generation timestamp

5 Production SQL DDL Migration Script

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-- Techvunex AI Assistant - Production DDL Migration
-- Compatible with: PostgreSQL 14+, MySQL 8.0+, SQLite 3.35+
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-- 1. Leads Table (Lead Generation Engine)
CREATE TABLE IF NOT EXISTS leads (
    id VARCHAR(36) PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(150),
    phone VARCHAR(30),
    company VARCHAR(150),
    service VARCHAR(100),
    requirement TEXT,
    budget VARCHAR(100),
    timeline VARCHAR(100),
    status VARCHAR(30) DEFAULT 'new',
    source VARCHAR(50) DEFAULT 'website_chat',
    human_required BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX IF NOT EXISTS idx_leads_email ON leads(email);
CREATE INDEX IF NOT EXISTS idx_leads_status ON leads(status);

-- 2. Contacts Table (Main Web Portal Sync)
CREATE TABLE IF NOT EXISTS contacts (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name VARCHAR(255),
    email VARCHAR(255),
    phone VARCHAR(50),
    company VARCHAR(255),
    subject VARCHAR(255),
    message TEXT,
    createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Users Table (Authentication)
CREATE TABLE IF NOT EXISTS users (
    id VARCHAR(36) PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(150) UNIQUE,
    phone VARCHAR(30),
    company VARCHAR(150),
    role VARCHAR(20) DEFAULT 'visitor',
    password_hash VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 4. Conversations Table (Chat Sessions)
CREATE TABLE IF NOT EXISTS conversations (
    id VARCHAR(36) PRIMARY KEY,
    session_id VARCHAR(100) UNIQUE NOT NULL,
    user_id VARCHAR(36),
    title VARCHAR(255) DEFAULT 'New Conversation',
    intent VARCHAR(50) DEFAULT 'general_question',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
);

-- 5. Messages Table (Dialogue Turns)
CREATE TABLE IF NOT EXISTS messages (
    id VARCHAR(36) PRIMARY KEY,
    conversation_id VARCHAR(36) NOT NULL,
    role VARCHAR(20) NOT NULL,
    content TEXT NOT NULL,
    sources JSON,
    token_usage JSON,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (conversation_id) REFERENCES conversations(id) ON DELETE CASCADE
);

-- 6. Feedbacks Table (Continuous Model Alignment)
CREATE TABLE IF NOT EXISTS feedbacks (

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    id VARCHAR(36) PRIMARY KEY,
    conversation_id VARCHAR(36) NOT NULL,
    message_id VARCHAR(36),
    rating INTEGER NOT NULL,
    feedback TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (conversation_id) REFERENCES conversations(id) ON DELETE CASCADE
);

-- 7. Documents Table (RAG Master Store)
CREATE TABLE IF NOT EXISTS documents (
    id VARCHAR(36) PRIMARY KEY,
    url VARCHAR(500) UNIQUE NOT NULL,
    title VARCHAR(255) NOT NULL,
    content TEXT NOT NULL,
    content_hash VARCHAR(64) NOT NULL,
    metadata_info JSON,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 8. Document Chunks Table (Neural Vector Embeddings)
CREATE TABLE IF NOT EXISTS document_chunks (
    id VARCHAR(36) PRIMARY KEY,
    document_id VARCHAR(36) NOT NULL,
    chunk_index INTEGER NOT NULL,
    content TEXT NOT NULL,
    embedding JSON,
    metadata_info JSON,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (document_id) REFERENCES documents(id) ON DELETE CASCADE
);

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